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Project Synopsis
on

Blockchain and ML-Based Pharmaceutical Supply Chain

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ABSTRACT

The pharmaceutical supply chain is a critical global infrastructure responsible for delivering safe, authentic medicines from manufacturers to patients. Despite blockchain technology's capability to create immutable, transparent transaction records, existing blockchain-based pharmaceutical supply chain systems lack intelligent fraud detection capabilities - relying exclusively on cryptographic traceability rather than predictive analytics. The World Health Organization estimates that counterfeit and substandard medicines account for up to 10% of the global drug supply, rising to 30% in developing nations, resulting in approximately 200,000 preventable deaths annually and an annual economic loss of USD 200 billion to legitimate pharmaceutical manufacturers.

This paper proposes a Blockchain and ML-Based Pharmaceutical Supply Chain, integrating Algorand smart contracts with three purpose-built ML modules: (1) Fraud and Anomaly Detection using Isolation Forest and XGBoost trained on blockchain-derived behavioural transaction features; (2) Counterfeit Medicine Risk Scoring using Random Forest to classify medicine batches as low, medium, or high risk from supply-chain metadata without requiring physical sample analysis; and (3) Inventory Demand Forecasting using LSTM and XGBoost Regression trained on blockchain-verified historical order data.

The central research question is: Do blockchain-derived behavioural features - including ownership transfer count, route deviation, timestamp irregularities, and verification failures - improve ML fraud and anomaly detection accuracy over conventional supply-chain features alone? This is empirically tested through a controlled experiment comparing Experiment A (traditional supply-chain features) against Experiment B (traditional features augmented with blockchain-derived behavioural features), evaluated using precision, recall, F1-score, and ROC-AUC. The system extends the foundational work of Abbas et al. (2020) - the most widely cited blockchain-ML pharmaceutical supply chain paper with 471 citations - by operationalising machine learning for active supply chain security rather than passive drug recommendation, addressing the gap those authors explicitly acknowledged as future work.

TABLE OF CONTENTS

Abstract.....	i
1. Introduction.....	1
2. Literature Survey / Initial Research	3
2.1 Review of Existing Literature.....	3
2.2 Literature Comparison Table.....	3
2.3 System Comparison Table.....	6
3. Problem Statement.....	8
4. Proposed Methodology.....	10
4.1 System Architecture.....	10
4.2 Module 1 - Fraud and Anomaly Detection.....	10
4.3 Module 2 - Counterfeit Medicine Risk Scoring.....	11
4.4 Module 3 - Inventory Demand Forecasting.....	12
4.5 Research Experiment Design.....	12
4.6 Technology Stack.....	13
4.7 System Workflow.....	14
5. Conclusions.....	15
6. References.....	16

1. INTRODUCTION

The global pharmaceutical industry is one of the most regulated and safety-critical supply chains in the world. From raw material procurement to final patient delivery, medicines pass through a complex network of suppliers, manufacturers, wholesalers, distributors, and pharmacies - each transition representing an opportunity for fraud, diversion, or counterfeit injection. The FBI and International Anti-Counterfeiting Coalition (IACC) have classified pharmaceutical counterfeiting as one of the largest criminal enterprises of the twenty-first century, with the problem growing substantially due to the rise of online pharmaceutical markets and the increasing sophistication of counterfeit producers.

Blockchain technology has emerged as a promising solution to the traceability and immutability challenges of pharmaceutical supply chains. By recording every transaction in a cryptographically secured, distributed ledger, blockchain ensures that the provenance and custody history of each medicine batch is permanently preserved and tamper-evident. Several recent studies - including Gomasta et al. (2023) and Dashtizadeh et al. (2022) - have demonstrated the feasibility of blockchain-based pharmaceutical supply chain management systems. However, a critical limitation persists across all such systems: the blockchain records what happened but cannot predict what is suspicious. Tamper detection is reactive, not proactive.

The most directly relevant prior work is that of Abbas et al. (2020), who proposed a blockchain and machine learning-based drug supply chain management and recommendation system (DSCMR), achieving 471 citations and representing the state of the art in combined blockchain-ML pharmaceutical systems. However, the machine learning component of DSCMR was used exclusively for drug recommendation based on patient reviews - not for fraud detection, anomaly identification, or counterfeit risk assessment. Abbas et al. themselves acknowledged this as a limitation, stating in their conclusion that "we will also improve our machine learning models in terms of accuracy" as future work. No subsequent paper has operationalised ML for active supply chain security in a blockchain-pharma context using transaction-derived behavioural features.

This paper addresses that gap by proposing a system with three ML modules operating directly on blockchain-derived data from an Algorand smart contract-based pharmaceutical supply chain:

- Module 1 - Fraud and Anomaly Detection: Isolation Forest (unsupervised) and XGBoost (supervised with synthetic anomaly injection) trained on blockchain-derived behavioural features such as ownership transfer count, route deviation score, timestamp irregularities, and verification failure count. This module provides real-time anomaly risk scores for every supply chain transaction.
- Module 2 - Counterfeit Medicine Risk Scoring: A Random Forest classifier that scores each medicine batch as Low, Medium, or High counterfeit risk using supply-chain metadata - supplier reliability history, batch size, manufacturing location, chain verification status, and distribution duration - without requiring physical sample analysis. This fundamentally differentiates the approach from all existing counterfeit detection literature, which requires a physical sample and image or spectroscopic analysis.
- Module 3 - Inventory Demand Forecasting: LSTM and XGBoost Regression models trained on blockchain-verified historical order data to predict medicine demand by region and time period, enabling proactive inventory management and reducing the risk of supply shortages that create conditions for counterfeit entry.

The overarching research question is: Do blockchain-derived behavioural features improve machine learning-based fraud and anomaly detection accuracy over conventional supply-chain features alone? The answer to this question - validated through a controlled A/B experiment with standard classification metrics - constitutes the primary measurable research contribution of this work.

2. LITERATURE SURVEY / INITIAL RESEARCH

2.1 Review of Existing Literature

A comprehensive literature review was conducted across four thematic areas directly relevant to the proposed system: (1) Blockchain-based pharmaceutical supply chain management with ML integration; (2) Anomaly and fraud detection in blockchain transaction environments; (3) Machine learning-based counterfeit medicine detection; and (4) Demand and inventory forecasting in pharmaceutical supply chains. The review covers 11 peer-reviewed papers published between 2020 and 2025, across IEEE Xplore, Elsevier, MDPI, and Springer venues, in addition to the baseline work by Abbas et al. (2020). The key findings are summarised in the comparison table below.

The baseline paper - Abbas et al. (2020), "A Blockchain and Machine Learning-Based Drug Supply Chain Management and Recommendation System for Smart Pharmaceutical Industry," Electronics (MDPI), 471 citations - represents the closest prior work to the proposed system. Their system (DSCMR) uses Hyperledger Fabric for blockchain-based supply chain management and applies N-gram, LightGBM, and sentiment analysis to the UCI Drug Review dataset for patient-facing drug recommendation. Critically, no ML is applied to supply chain transactions for fraud detection, anomaly scoring, or counterfeit risk prediction. The authors explicitly identify improving ML models as future work, establishing the precise gap this paper targets.

2.2 Literature Comparison Table

#	Authors & Year	Title / Venue	ML Technique	Key Result / Metric	Gap Addressed by Our Work
B	Abbas et al. (2020) Electronics, MDPI	A Blockchain and ML-Based Drug SCM and Recommendation System	N-gram + LightGBM + Sentiment Analysis (drug recommendation)	System functional; ML used only for patient-facing recommendation; no	Baseline: we extend their ML layer to active supply-chain security instead of drug

				fraud/anomaly ML	recommendation
1	Gomasta et al. (2023) Heliyon, Elsevier	PharmaChain: Blockchain-based drug supply chain provenance verification system	None - ECDSA signatures + SHA-256 only	Tamper-evidence verified; resilient to 33% malicious-node attack	No ML intelligence; cannot predict or score suspicious transactions before an incident occurs
2	Dashtizadeh et al. (2022) IEEE MysuruCon	Secure Blockchain-based Pharmaceutical SCM: Traceability and Detection of Counterfeit Covid-19 Vaccines	Blockchain+ML demand forecasting proposed conceptually; no model trained or evaluated	Architecture proposed; ML left as future work; no accuracy metrics reported	ML integration stated as intention only - we operationalise it with three trained, evaluated modules
3	Gazzola et al. (2023) Sustainability, MDPI	Using the Transparency of Supply Chain Powered by Blockchain to Improve Sustainability Relationships with Stakeholders in the Food Sector: The Case Study of Lavazza	Smart contract automation; no ML component	Confirms Algorand (Pure Proof-of-Stake) as production-ready platform for supply chain traceability	Confirms our platform choice; no ML added - our system addresses this gap

4	Hasan et al. (2024) Blockchain: Research and Applications, Elsevier	Detecting Anomalies in Blockchain Transactions using ML Classifiers and Explainability Analysis	XGBoost with XGBCLUS under-sampling + SHAP explainability	Improved illicit Bitcoin transaction detection under class imbalance	Domain = cryptocurrency; no pharma-specific features; no supply-chain custody concepts
5	Ashfaq et al. (2022) Sensors, MDPI	A Machine Learning and Blockchain Based Efficient Fraud Detection Mechanism	XGBoost + Random Forest on blockchain transactions	Effective Bitcoin fraud classification; high precision and AUC reported	Financial domain only; no route deviation, batch identity, or custody-transfer features
6	Wan (2021) IEEE ICBAIE	XGBoost Based Supply Chain Fraud Detection Model	Hybrid XGBoost + Random Forest	Strong fraud classification on DataCo e-commerce dataset (~180K records)	General e-commerce; no blockchain involvement; no pharma-specific domain features
7	Jumani & Raza (2025) Computers, MDPI	ML for Anomaly Detection in Blockchain: A Critical Analysis, Empirical Validation, and Future Outlook	Benchmark: RF, AdaBoost, XGBoost, K-means, Isolation Forest	RF and XGBoost highest accuracy for blockchain network anomalies (51% attacks)	Network-level anomalies only; no application-level or supply-chain domain; validates our model choices
8	Thomson & Varuna (2024) IC3T 2023, Springer	Analyzing the Counterfeit Medicines Based on Classification	Random Forest, Naive Bayes, SVM on medicine images	SVM achieves 94.5% accuracy for counterfeit image	Requires physical sample image; cannot score risk from

	LNNS Vol. 897, pp.355-361	Using ML Techniques		classification	supply-chain metadata before physical inspection
9	Parvathi et al. (2025) MIJARCSE, Vol. 11, Issue 1 (2025)	Blockchain-Enabled Product Tokenization and CNN-Based Counterfeit Detection for Secure Supply Chain Verification	CNN on general product images (Roboflow fake-product dataset) + Ethereum ERC-721 token verification	Image-based CNN detection; most recent paper still requires physical sample	Cross-domain CNN counterfeit system - still needs a physical sample, not metadata-based
10	Zhu et al. (2021) Production and Operations Management	Demand Forecasting with Supply-Chain Information and ML: Evidence in the Pharmaceutical Industry	Cross-series ML framework using downstream inventory information as features	Incorporating supply-chain info materially improves forecast accuracy	No blockchain; assumes centrally available trusted data - we ground forecasts in verified ledger data
11	Yani & Aamer (2023) Int. J. Pharmaceutical and Healthcare Marketing	Demand Forecasting Accuracy in the Pharmaceutical SCM: A ML Approach	Random Forest and decision-tree models; 80/20 train-test split	RF outperforms traditional forecasting by 10-41%; optimal split identified	Single-case study; no blockchain; no fraud/counterfeit scope; we extend with ledger-verified inputs

2.3 System Comparison Table

The table below compares the proposed system against the most closely related existing systems across key functional and research dimensions:

Feature / Capability	Proposed System	Abbas et al. 2020	PharmaChain 2023	Hasan et al. 2024	Jumani & Raza 2025
Blockchain-based SCM	✓	✓	✓	✓	✗
Algorand Smart Contracts	✓	✗	✗	✗	✗
ML Fraud/Anomaly Detection	✓	✗	✗	Crypto only	Crypto only
Pharma-Specific ML Features	✓	✗	✗	✗	✗
Counterfeit Risk Scoring (metadata)	✓	✗	✗	✗	✗
Inventory Demand Forecasting	✓	✗	✗	✗	✗
Blockchain-Verified ML Inputs	✓	✗	✗	Partial	✗
A/B Feature Experiment	✓	✗	✗	✗	✗
Explainability (SHAP/Feature Importance)	✓	✗	✗	✓	✗
Admin Intelligence Dashboard	✓	✗	✗	✗	✗

As demonstrated in the comparison, no existing system combines blockchain-native pharmaceutical supply chain data with a multi-module ML layer performing fraud detection, counterfeit risk scoring, and demand forecasting using pharma-specific behavioural features. This constitutes the primary and unaddressed research gap that the proposed system targets.

3. PROBLEM STATEMENT

Analysis of the existing literature and operational pharmaceutical supply chain systems identifies five critical, interconnected problems that the proposed system is designed to address:

1. **The Reactive Traceability Problem:** Existing blockchain-based pharmaceutical supply chain systems - including the most sophisticated such as PharmaChain (Gomasta et al., 2023) - create immutable records of every transaction but operate entirely in retrospect. The blockchain can confirm what happened after a fraudulent or counterfeit transaction is discovered, but provides no mechanism for predicting, scoring, or flagging suspicious events proactively. This reactive posture means that counterfeit medicines may circulate for extended periods before detection.
2. **The ML-Blockchain Disconnection Problem:** The sole prior work combining ML with blockchain pharmaceutical supply chains - Abbas et al. (2020) - applies machine learning exclusively to patient-facing drug recommendation using public drug review data. No existing system uses blockchain transaction records as ML training features. This disconnection means the rich behavioural signals encoded in the blockchain - transfer frequency, route patterns, timestamp distributions, ownership hop counts - are entirely unused for security intelligence.
3. **The Physical Inspection Bottleneck for Counterfeit Detection:** All existing ML-based counterfeit medicine detection systems - including the most recent work by Thomson and Varuna (2024) and Parvathi et al. (2025) - require a physical medicine sample for image classification or spectroscopic analysis. This means suspicious batches can only be identified after they have already entered the distribution chain and reached an inspection point. No system currently scores counterfeit risk from supply-chain provenance metadata alone, before physical inspection is possible.
4. **The Absence of a Measurable Research Contribution in Blockchain-Pharma ML:** Existing blockchain-pharma papers that discuss ML integration treat it as a stated future direction without delivering experimental results. No paper has formulated and empirically tested the hypothesis that blockchain-derived behavioural features improve ML model performance for supply chain fraud detection - a gap this research

addresses through a controlled A/B experimental design with standard classification metrics.

5. The Demand-Security Disconnection: Pharmaceutical demand forecasting research (Zhu et al., 2021; Yani and Aamer, 2023) is methodologically mature but entirely decoupled from blockchain security systems. Demand forecasting currently uses centrally reported, unverified figures; supply shortages created by inaccurate forecasts generate conditions that counterfeit producers exploit. Integrating blockchain-verified demand signals with ML forecasting creates a security-aware inventory management capability that no existing work provides.

The broad aim of the proposed work is to develop, implement, and empirically evaluate a unified blockchain and machine learning framework that simultaneously resolves all five identified problem dimensions within a single, cohesive pharmaceutical supply chain management system.

4. PROPOSED METHODOLOGY

4.1 System Architecture

The proposed system follows a dual-layer architecture comprising a Blockchain Supply Chain Management Layer and a Machine Learning Intelligence Layer, integrated via a Python-based REST API microservice. The Blockchain Layer - built on Algorand smart contracts (PyTeal / Algorand Python) deployed via the Algorand SDK, ASP.NET (C#) backend, and Microsoft SQL Server (MSSQL) - handles all supply chain transaction recording, role-based access control, and immutable provenance management. The ML Layer - a Python Flask/FastAPI service - reads transaction data from MSSQL, applies the three trained ML models, and returns risk scores and forecasts to the ASP.NET application for display on the administrator intelligence dashboard.

The six supply chain participant roles - Admin, Supplier, Manufacturer, Wholesaler, Distributor, and Customer - interact with the system through role-specific web portals. Every transaction (raw material order, production batch creation, custody transfer, delivery confirmation) is recorded on the Algorand blockchain via smart contract execution and simultaneously persisted to MSSQL. The ML service reads this dual-stored data to generate real-time intelligence outputs.

4.2 Module 1 - Fraud and Anomaly Detection

This module constitutes the primary research contribution of the system. Two complementary models are trained:

- Isolation Forest (Unsupervised): Trained on normal transaction behaviour to assign an anomaly score to each new transaction without requiring labelled fraud examples. Isolation Forest is selected because, as empirically validated by Jumaní and Raza (2025), it is among the highest-performing unsupervised algorithms for blockchain transaction anomaly detection.
- XGBoost Classifier (Supervised): Trained on a dataset comprising legitimate transactions from the Kaggle Blockchain-Based Medicine Supply Chain Management dataset augmented with synthetically injected anomalous transactions. Anomalies are introduced according to predefined fraud patterns derived from known supply chain

attack vectors: unexpected ownership transfers, route deviations, unusually short transfer intervals, and quantity discrepancies. This methodology follows the approach validated by Hasan et al. (2024).

The feature set for both models spans two categories - the comparison of which constitutes the A/B experiment:

- Experiment A - Traditional Supply-Chain Features: `supplier_id`, `manufacturer_id`, `medicine_category`, `batch_size`, `manufacturing_location`, `distribution_duration_days`, `quantity_ordered`, `quantity_received`, `quantity_discrepancy`.
- Experiment B - Traditional + Blockchain-Derived Behavioural Features: all Experiment A features plus `transfer_count` (total ownership transfers), `ownership_hops` (number of intermediate entities), `route_deviation_score` (deviation from registered route), `timestamp_gap_hrs` (time between consecutive transfers), `verification_fail_count` (failed blockchain verification attempts), and `unexpected_entity_flag` (binary: unknown participant in chain).

Models trained on Experiment B features are expected to demonstrate measurably higher precision, recall, F1-score, and ROC-AUC than those trained on Experiment A features alone. This comparison constitutes the empirical research contribution of the module.

4.3 Module 2 - Counterfeit Medicine Risk Scoring

This module addresses the most critical gap identified in the literature review: all existing counterfeit detection ML systems require a physical medicine sample. Module 2 scores counterfeit risk from supply-chain provenance metadata alone, enabling proactive flagging of suspicious batches before physical inspection.

A Random Forest classifier is trained to classify each medicine batch as Low Risk (0-35%), Medium Risk (36-70%), or High Risk (71-100%) for counterfeit status. The training dataset is constructed by deriving risk labels from simulated supply chain scenarios where known counterfeit injection patterns (unregistered suppliers, unusually short manufacturing times, route bypassing, multiple unexpected ownership transfers) are used to assign ground-truth risk levels. Features include: `supplier_reliability_score` (derived from historical complaint and delivery records), `batch_size`, `manufacturing_location_risk_index`, `chain_verification_status`

(proportion of transfers verified on blockchain), transfer_count, unexpected_transfer_flag, distribution_duration_deviation, and complaint_history_count.

The output - a probability score and categorical risk label (Low/Medium/High) - is displayed in the admin dashboard alongside the specific feature contributions (SHAP values) explaining the risk score, ensuring the system is interpretable and actionable for supply chain administrators.

4.4 Module 3 - Inventory Demand Forecasting

This module extends the pharmaceutical demand forecasting literature of Zhu et al. (2021) and Yani and Aamer (2023) by grounding predictions in blockchain-verified order history rather than centrally reported figures, making the forecasts themselves resistant to the data manipulation that Module 1 is designed to detect elsewhere in the system.

Two models are evaluated: an LSTM (Long Short-Term Memory) recurrent neural network for capturing long-range seasonal patterns, and an XGBoost Regression model for interpretable feature-based forecasting. Training features include: medicine_id, region, month, historical_demand (12-month rolling window from blockchain-verified orders), seasonal_flag, complaint_rate (from blockchain records), batch_expiry_proximity_days, and previous_stockout_flag. The model with superior Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) on the held-out test set is selected for deployment. Forecast outputs are displayed in the admin dashboard with current stock levels and projected demand for the next 30 days, enabling proactive restocking decisions.

4.5 Research Experiment Design

The primary research hypothesis is: H1 - Blockchain-derived behavioural features (transfer count, route deviation, timestamp irregularities, ownership hops, verification failures) improve ML-based fraud and anomaly detection accuracy in pharmaceutical supply chains compared with conventional supply-chain features alone. The null hypothesis H0 states that there is no statistically significant difference in detection performance between Experiment A (conventional features) and Experiment B (conventional + blockchain-behavioural features).

Evaluation metrics across both experiments: Precision, Recall, F1-Score (macro-averaged), ROC-AUC, and Average Precision. Models tested: Isolation Forest, XGBoost, and Random

Forest. If Experiment B models demonstrate statistically significant improvement over Experiment A counterparts on F1-Score and ROC-AUC (tested via McNemar's test, $\alpha = 0.05$), H1 is supported, providing empirical evidence that blockchain-derived behavioural features carry meaningful discriminative signal for pharmaceutical supply chain fraud detection.

4.6 Technology Stack

Component	Technology
Blockchain Platform	Algorand Blockchain (TestNet / MainNet)
Smart Contracts	Algorand Smart Contracts (PyTeal / Algorand Python) + py-algorand-sdk
Cryptographic Hashing	SHA (Secure Hash Algorithm)
Backend Framework	ASP.NET with C#
Database	Microsoft SQL Server (MSSQL)
Frontend	HTML, CSS, JavaScript
ML Service	Python 3.11, Flask / FastAPI (REST API)
ML Libraries	scikit-learn (Isolation Forest, RF, XGBoost), TensorFlow / Keras (LSTM)
Explainability	SHAP (SHapley Additive exPlanations)
Data Processing	pandas, NumPy
IDE	Microsoft Visual Studio
ML Training Environment	Jupyter Notebook / Google Colab
Dataset	Kaggle: Blockchain-Based Medicine Supply Chain Management (Bidyutmala Saha) + synthetic anomaly injection

4.7 System Workflow

1. Admin registers all supply chain participants (Supplier, Manufacturer, Wholesaler, Distributor, Customer) via the web portal; smart contracts created and deployed on the Algorand network via the Algorand SDK.
2. Supplier logs raw material stock; Manufacturer places raw material orders. Each transaction recorded as a blockchain block via smart contract execution and simultaneously stored in MSSQL.
3. Manufacturer creates medicine batches, records production details on blockchain; Wholesaler and Distributor place orders; each custody transfer logged as a smart contract transaction.
4. Customer verifies medicine authenticity by scanning QR code; system retrieves blockchain provenance record and displays full supply chain history.
5. ML Service reads transaction records from MSSQL; Module 1 computes anomaly scores for each transaction in real time; Module 2 scores counterfeit risk for each new batch; Module 3 updates demand forecasts daily.
6. Admin Intelligence Dashboard displays: real-time anomaly alerts with SHAP-based explanations, counterfeit risk heatmap by batch and supplier, demand forecast vs. current stock levels, and supply chain health overview.
7. High-risk transactions (anomaly score > 0.80 or counterfeit risk = High) trigger automatic admin notification for manual investigation prior to approving distribution.

5. CONCLUSIONS

This synopsis has presented a Blockchain and ML-Based Pharmaceutical Supply Chain - a framework that addresses five critical problems in pharmaceutical supply chain management: reactive traceability, the ML-blockchain disconnection, the physical inspection bottleneck for counterfeit detection, the absence of a measurable ML research contribution, and the demand-security disconnection.

The proposed system makes four distinct research contributions. First, it uses blockchain-derived behavioural transaction features - ownership transfer count, route deviation, timestamp irregularities, and verification failures - as ML training inputs for pharmaceutical supply chain fraud detection. Second, Module 2 introduces metadata-based counterfeit risk scoring without requiring a physical medicine sample. Third, the controlled A/B experiment comparing detection performance with and without blockchain-derived behavioural features provides an empirically verifiable research hypothesis. Fourth, demand forecasting is grounded in blockchain-verified transaction history, making forecasts more resistant to data manipulation.

The system extends the baseline of Abbas et al. (2020) by completing the machine learning expansion they identified as future work, replacing passive drug recommendation with active, multi-dimensional supply chain intelligence. The practical significance extends across stakeholders: patients benefit from a higher-integrity medicine supply, manufacturers reduce counterfeiting losses, regulatory bodies gain a proactive monitoring tool, and researchers receive a framework for blockchain-behavioural feature engineering.

Future work will explore IoT temperature and humidity sensors for cold-chain monitoring, federated learning for cross-institutional model training, and natural language processing of regulatory inspection reports as additional counterfeit risk signals.

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